MMAFC Proposal for Inventory Management Application

Goals

MyMobileAppFictionCompany (MMAFC) proposes the following mobile inventory application for the clients to manage both their inventory and user login credentials. The initial purpose of this application is for internal users to be able to log in securely and view and manage inventory stored in a warehouse. The main features of the application will be displaying all items in the inventory in a grid format, the ability for users to add, delete, increase, and decrease items from the inventory, and a notification system for when an inventory item has been reduced to zero.

Users

At the outset, three initial user types have been identified based on the initial version of this application. The notification system alerting all users when stock has been reduced to zero is a critical part for each of the users, as it may prompt reordering, notification to external clients, changes in monthly reporting, or fiscal planning, etc.

1) Small Business Owners. These users may have a small number of staff who need to review and adjust inventory in real-time to make ordering decisions, fill orders from external clients, and create auditing reports.
2) Education/Office Administration. These users may need to maintain a supply management system for routine office supplies, and standard school-based supplies for running a school, such as toilet paper, lunchroom food and supplies, to plan for fiscal and monthly ordering, and to create auditing reports.

3) Warehouse Staff. These users would likely do the bulk of the increasing and decreasing inventory as they move items throughout the warehouse, either as small business employees, office supply management, or educational supply warehouse staff.

Optional Considerations

Though not necessarily the intended audience, as the application is intended to align with a warehouse, home/personal users may also see benefit. For example, household members could review the inventory list comprised of household goods while shopping to see low-stock and out-of-stock items, or as a hobbyist managing craft materials or collections.

An additional consideration to increase the user benefit of the application is to incorporate the ability to create a custom notification, such as "low stock parameters." This enhanced notification option would increase the personalization for users and competitiveness of the application, and allow for better inventory management planning.

UI Design

Screens and Features

On application launch, the user will see a standard login credential screen with the buttons for <Login> and <New User> for application authentication. Once the application is open, the main tab <Home> is displayed with four navigation buttons along the bottom for <Home>, <Notifications>, and <Logout>. <Home> will have the inventory displayed in grid format. On this screen, users can manage individual inventory item sizing. Users will be able to delete items from the inventory, and there will be a card with a <+> symbol for adding a new inventory item. The notifications tab displays any new notifications at the top of the screen, followed by old notifications in reverse chronological order. The logout tab displays a standard dialog pop-up to confirm logout, yes or no. A future consideration is to incorporate a settings tab

that would display options such as dark mode, font sizing, and grid sizing for a better user experience and accessibility considerations.

Design Considerations

In the review of *UI and Content for Billions* (Android Developers, 2024) and Material Design's guidelines on accessibility, the following design considerations will be incorporated. "A placeholder UI [used at] launch screen to reduce the perceived loading time" (Android Developers, 2024). This simply may be a <Hello> text during loading. This concept is also recommended as some older devices or operating systems may take longer for the application to launch and communicate with the user that the application is functioning as intended. There will be haptic "feedback for all touchable items" (Android Developers, 2024). This is intended "to reduce the perceived app latency" (Android Developers, 2024) but is also a recommended accessibility function to communicate to the users that the application has registered their interaction (Material Design, n.d.). Additionally, "Standard platform dialog" confirmation boxes will be utilized whenever possible as part of the continuity of user communication, and color scheme selection controls will be applied (Material Design, n.d.).

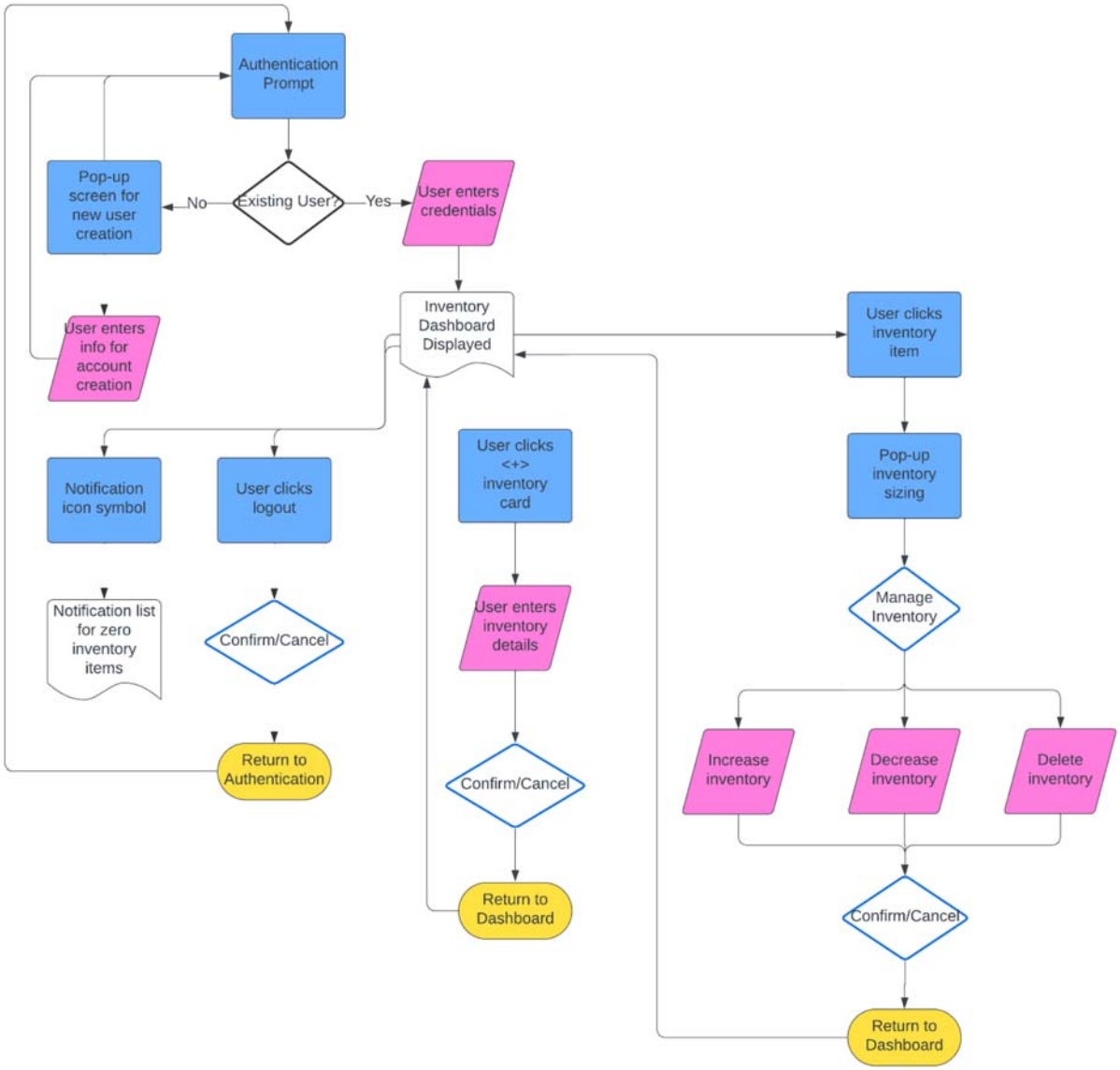
User Navigation

Overall, the user will navigate through screens in four ways. Users can 1) directly navigate by tapping the screen with the haptic feedback; 2) use labeled tabs at the bottom of the screen; 3) use gesture navigation to scroll through the inventory list; and 4) interact with dialog navigation through pop-up screens and confirmations.

Upon application launch, the login screen will contain entry boxes for user credentials and password with the widgets for <Login> or <New User?>. When the user clicks the New User widget, a standard platform dialog box will appear with a form to create a new user for the

application. Once authenticated, the user will be brought to the main application tab. When the user selects an inventory item from the grid, the item will be highlighted in color and a standard dialog box pop-up will display with the option to increase or decrease the quantity, as a plus or minus symbol on either side of the quantity number, a standard garbage can icon, and then buttons for <Confirm> or <Cancel>; either selection closes the pop-up box and returns the user to the inventory grid. If a user wants to add an item to the inventory, there is a card with a <+> symbol at the top of the inventory items grid to do so. When selected, a new screen will open with a form for the item's name and quantity. The user can scroll through the inventory list by swiping the screen. When a user clicks the notification icon, the notifications will be displayed in reverse chronological order. Notifications should be highlighted when new and lose the highlighting after the user has read the message.

Draft Process Flow. Below is a draft process flow for user navigation.



Code Design

Functional Requirements

The application requires a database with two tables: one for the inventory and one for the user credentials. An evaluation of data caching and synchronization mechanisms needs to occur to ensure the UI is up to date, but keeps latency in mind (Android Developers, 2024).

User Credentials Table

On the login screen, users will enter their username and password to authenticate against the user credentials data table. A pop-up message should occur when that username does not exist and prompt for a new user account. When a user does not have an account, they will create one by filling out the required elements on the new user form, which gets saved to the database. The new user's credentials are authenticated against the updated data table during login.

Inventory Table

The inventory table contains all the inventory items displayed on the <Home> screen, which is populated after user authentication. Most of the data contained in this table is generated by the user via the UI components for increasing and decreasing item quantities, as well as adding and deleting inventory items. All inventory modifications will have <Confirm> and <Cancel> buttons to complete these actions. When users input new or otherwise adjust inventory and confirm the change, the data entry corresponds to backend operations adjusting the table in the database. The table then returns those modified values/inventory changes as updated on the <Home> screen. A refresh widget should be added to initiate a call to the database for updated information. The initial design is for inventory item details to include name and quantity. Future releases may incorporate short descriptions and photos to the item description.

Notifications

There are three ways in which notifications are triggered for the application. 1) Upon launching the application, the UI requests an update of the inventory list. During this update, should the backend/database detect that an inventory item quantity is zero, a notification will pop up, and the icon at the bottom of the screen will have an indicator (dot or number) to also alert the user that there is an inventory notification. On the pop-up, the user will have a button to

acknowledge, and the pop-up will dismiss. 2) The user clicks the refresh button on the inventory list <Home> screen, which invokes #1. 3) A user decreases the quantity of an item to zero during inventory management, which triggers #1 upon confirmation of the quantity adjustment. If one user reduces an inventory item to zero and if more than one user is interacting with the same database, all users currently utilizing the application will receive the notification. Future releases of the application could create push notifications from the application about zero-quantity items.

References

Android Developers. (2024). UI and Content for Billions. Google for Developers. Retrieved from: https://developer.android.com/docs/quality-guidelines/build-for-billions/ui

Material Design. (n.d.). Foundations. Google. Retrieved from: https://m3.material.io/foundations